A LOCAL RING WITH A UNIQUE PRIME IDEAL AND A NONZERO FLAT IDEAL

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Context span 2851-2861: R = k[X_i, Y_i]/(X_i - Y_i X_{i+1}, Y_i^2) with images x_i, y_i; the chain x_i = y_i x_{i+1} = y_i y_{i+1} x_{i+2} = ...; R has only one prime ideal, namely m = (x_i, y_i); the ideal I = (x_i) is claimed flat
- Context span 2863-2876: the annihilator of x_i in R is the ideal (x_1, x_2, ..., y_i, y_{i+1}, ...); the module M generated by e_i with relations e_i = y_i e_{i+1} is flat as the colimit of copies of R with transition maps R –y_1–> R –y_2–> R –y_3–> ... (colimits of flat modules with flat... the displayed transition system); the annihilator of e_i in M equals the annihilator of x_i in R; every element of M, respectively I, is f e_i, respectively h x_i, so the map M -> I sending e_i to x_i is an isomorphism and I is flat

nonzero flat ideal. Here is the construction. Let k be a field. Let X_i, Y_i, $i \geq 1$ be variables. Take $R = k[X_i, Y_i]/(X_i - Y_iX_{i+1}, Y_i^2)$. Denote x_i, resp. y_i the image of X_i, resp. Y_i in this ring. Note that

$$x_i = y_ix_{i+1} = y_iy_{i+1}x_{i+2} = y_iy_{i+1}y_{i+2}x_{i+3} = \ldots$$

in this ring. The ring R has only one prime ideal, namely $\mathfrak{m} = (x_i, y_i)$. We claim that the ideal $I = (x_i)$ is flat as an R-module.

Note that the annihilator of x_i in R is the ideal $(x_1, x_2, x_3, \ldots, y_i, y_{i+1}, y_{i+2}, \ldots)$. Consider the R-module M generated by elements e_i, $i \geq 1$ and relations $e_i = y_ie_{i+1}$. Then M is flat as it is the colimit $\operatorname{colim}_i R$ of copies of R with transition maps

$$R \xrightarrow{y_1} R \xrightarrow{y_2} R \xrightarrow{y_3} \ldots$$

Note that the annihilator of e_i in M is the ideal $(x_1, x_2, x_3, \ldots, y_i, y_{i+1}, y_{i+2}, \ldots)$. Since every element of M, resp. I can be written as fe_i, resp. hx_i for some $f, h \in R$ we see that the map $M \to I$, $e_i \to x_i$ is an isomorphism and I is flat.

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Lemma 0.1. *There exists a local ring R with a unique prime ideal and a nonzero ideal $I \subset R$ which is a flat R-module*

Proof. See discussion above. □